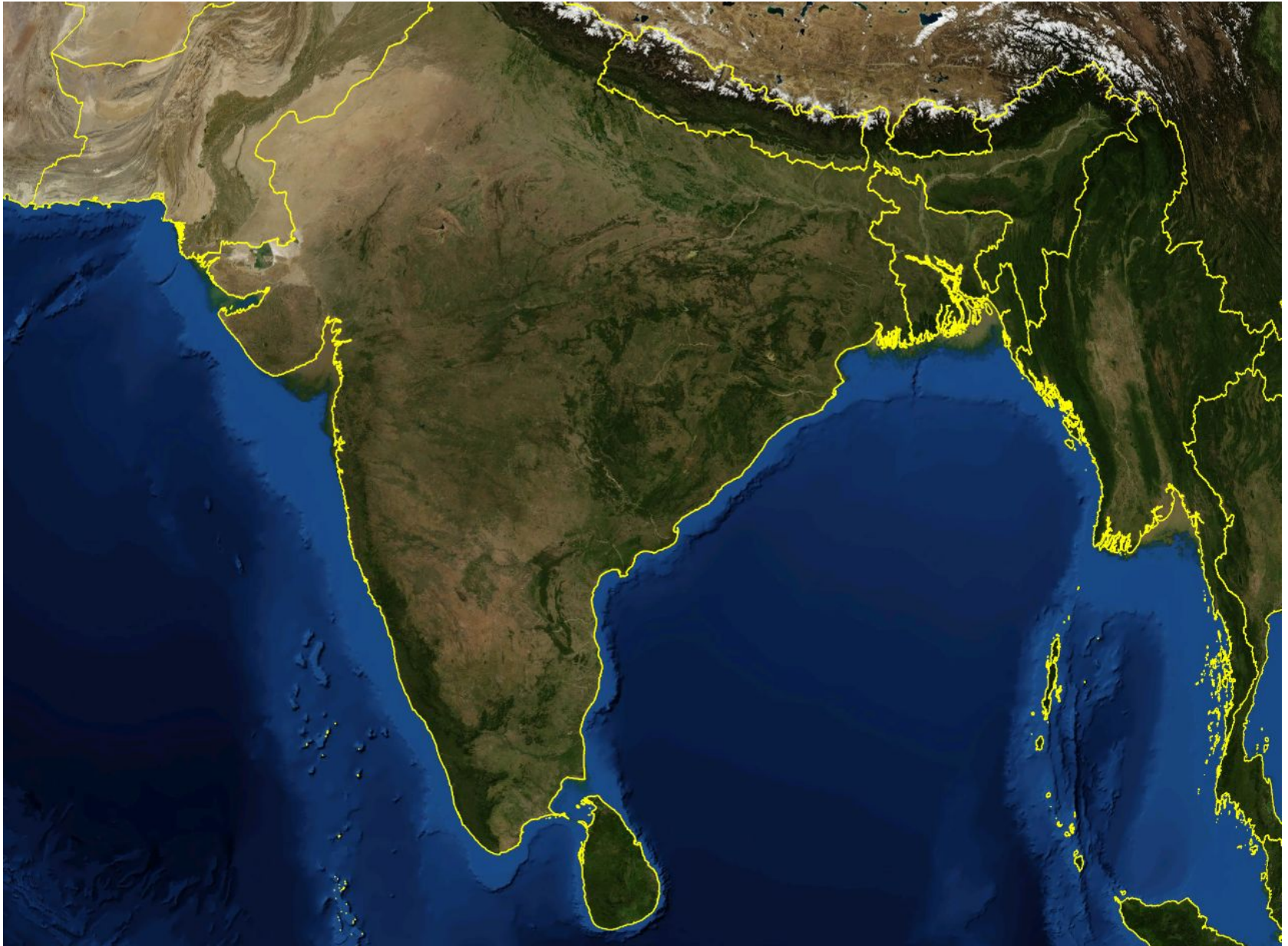


TOPIC 9: Environmental Problems of Bangladesh



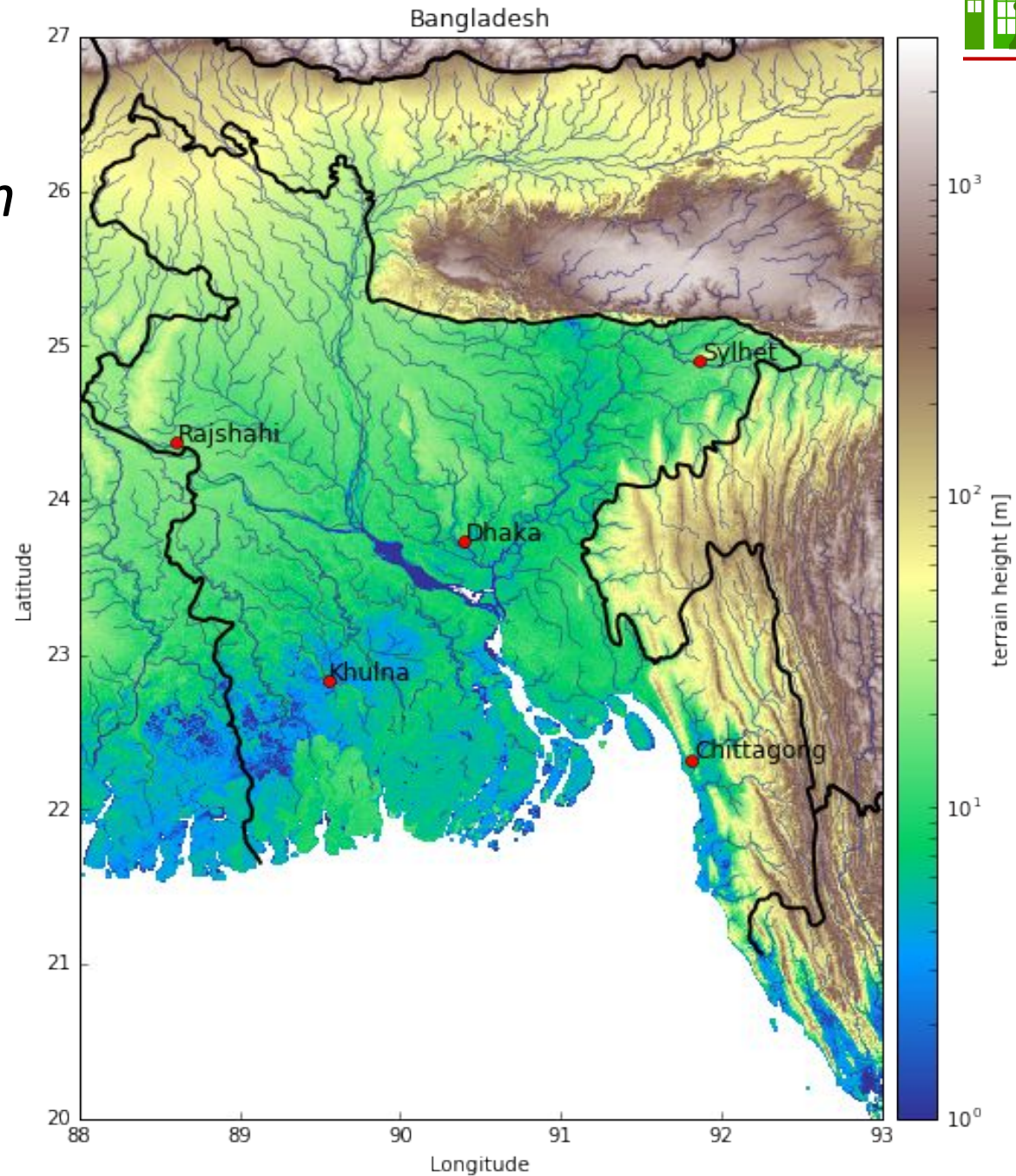
Geographic Location





Digital Elevation Model

Slope
Direction





Key Environmental Issues in Bangladesh:-

- Land Degradation
- Water Pollution and Scarcity
- Air Pollution
- Biodiversity
- Natural Disasters

Land Degradation



Cause(s)

✓ **human economic activities**

Key Environmental Issues ●●●

- ***Improper Cultivation in Hill Slopes, Terrace Land and Piedmont Plains***



- ***Faulty Irrigation***
 - impermeable highlands
 - cropping pattern
 - continued oxygen deprivation



Land Degradation(2)

- ***Imbalanced Fertilizer Use***

- ***Plough pan***

Soil drainage

restricts root penetration

movement of soil moisture

- ***Improper Use of Pesticides***

destroy useful topsoil

microbes





Land Degradation(3)

Key Environmental Issues...

- *Over Exploitation of Biomass from the Agricultural Fields*
- *Unplanned Rural Infrastructure (Road, Embankment, FCD/I)*
- *Urbanization*
- *Brickfields and Biomass Use*
- *Unplanned Industrial Development*



Land Degradation (4)

- *Mining of Sand and Gravels from Agricultural Land*
- *Land Ownership and Tenure*
- *deforestation*

Cause(s)
✓

Waterborne Land Degradation

- *Riverbank Erosion and Sedimentation*
- *Salinity*



WATER POLLUTION AND SCARCITY

Key Environmental Issues ● ● ●

✓ Causes of Water Pollution

- **Industrial effluent**
- **Agrochemical**
- **Fecal Pollution**
- **Oil and Lube Spillage**
- **Low Flow in Dry season**





WATER POLLUTION AND SCARCITY

✓ **Causes of Scarcity**

Key Environmental Issues ●●●



- *Upstream Withdrawal and Diversion of Flow*
- *Withdrawal of Wetland Water*
- *Gradual Siltation in River Bed and Floodplain*
- *Dry Season Irrigation*
- *Floods*
- *Dry Season Rainfall*



Causes **AIR POLLUTION**

- **Rapid Urbanization**
- **Emissions from Vehicles**
- **Unplanned Industrial Development**
- **Brick Kilns**

Key Environmental Issues ●●●





BIODIVERSITY

✓ Habitat Depletion and Over Exploitation

Key Environmental Issues...

Protected Areas defined under Bangladesh Wildlife Preservation Act, 1974

A. National Park: *a comparatively large area of natural beauty to which the members of the public have access for recreation, education and research, and in which the wildlife is protected.*

B. Wildlife Sanctuary: *an area maintained as an undisturbed breeding ground for wild fauna and where the habitat is protected for the continued well-being of the resident or migratory fauna.*

C. Game Reserve: *normally comprises a relatively isolated area meant for protection of wildlife in general and to increase the population of specified species.*



NATURAL DISASTERS

Key Environmental Issues...

- Geographical Setting and Physiography
- Hydrology
- **Global Environmental Pressures**
- ✓ 57 trans-boundary rivers; upstream deforestation, Heavy rainfall, melting of glaciers, and soil erosion, upstream withdrawal of water & Global Emission



Major Environmental Concerns

1. Agricultural Resource Base

✓ Over 60 per cent of the total land area in Bangladesh is cultivated. This is one of the highest percentages in Asia.

2. Biomass

3. Chemical Fertilizers and Pesticides

4. Pollution

5. Deforestation

6. Wetlands and Fisheries

7. Mangrove Ecosystem

8. Salinity

9. Health and Sanitation

10. Urbanization

11. Water Pollution

12. Global Warming and Climate Change



Climate Profile of Bangladesh

- ❑ Climate Vulnerability - It ranked first on German Watch's Global Climate Risk Index of 2011.

vulnerability is caused by a combination of biophysical factors (being a flat, low, delta country, exposed to flooding and cyclones) and socio-economic factors (such as high dependence on agriculture, population density, and poverty)



Temperatur

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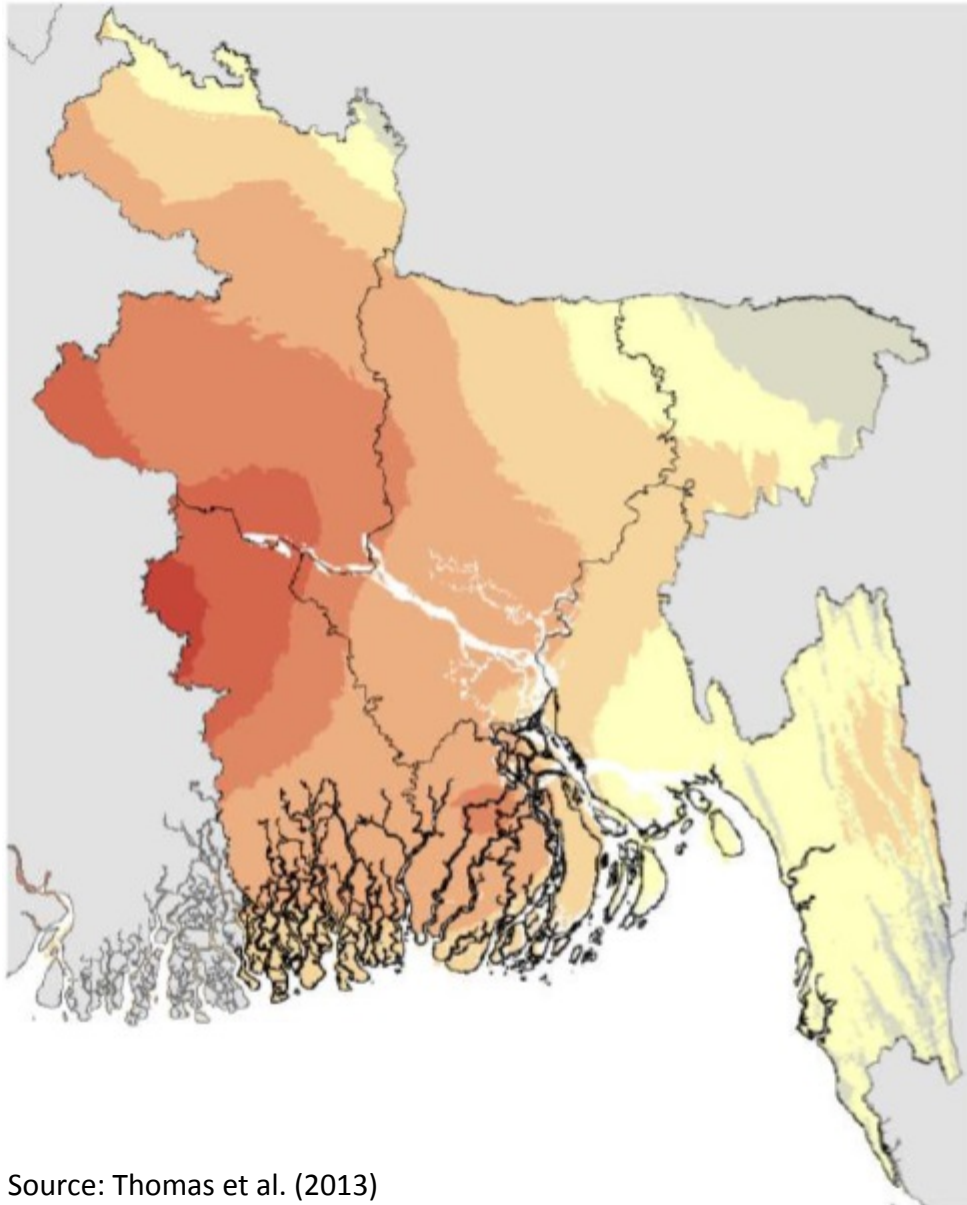
Bangladesh has a monsoon-type climate. It has three seasons:

- ❑ Hot, humid summers (March-June) with average maximum temperatures of 37°C;
- ❑ Cooler monsoon seasons (June-September);
- ❑ Dry, cooler winters (October-March) with average maximum temperatures of 28°C.

The highest temperatures are in the southwest, the lowest in the northeast of the country



Current average maximum temperature (1950-2000) in degrees Celsius

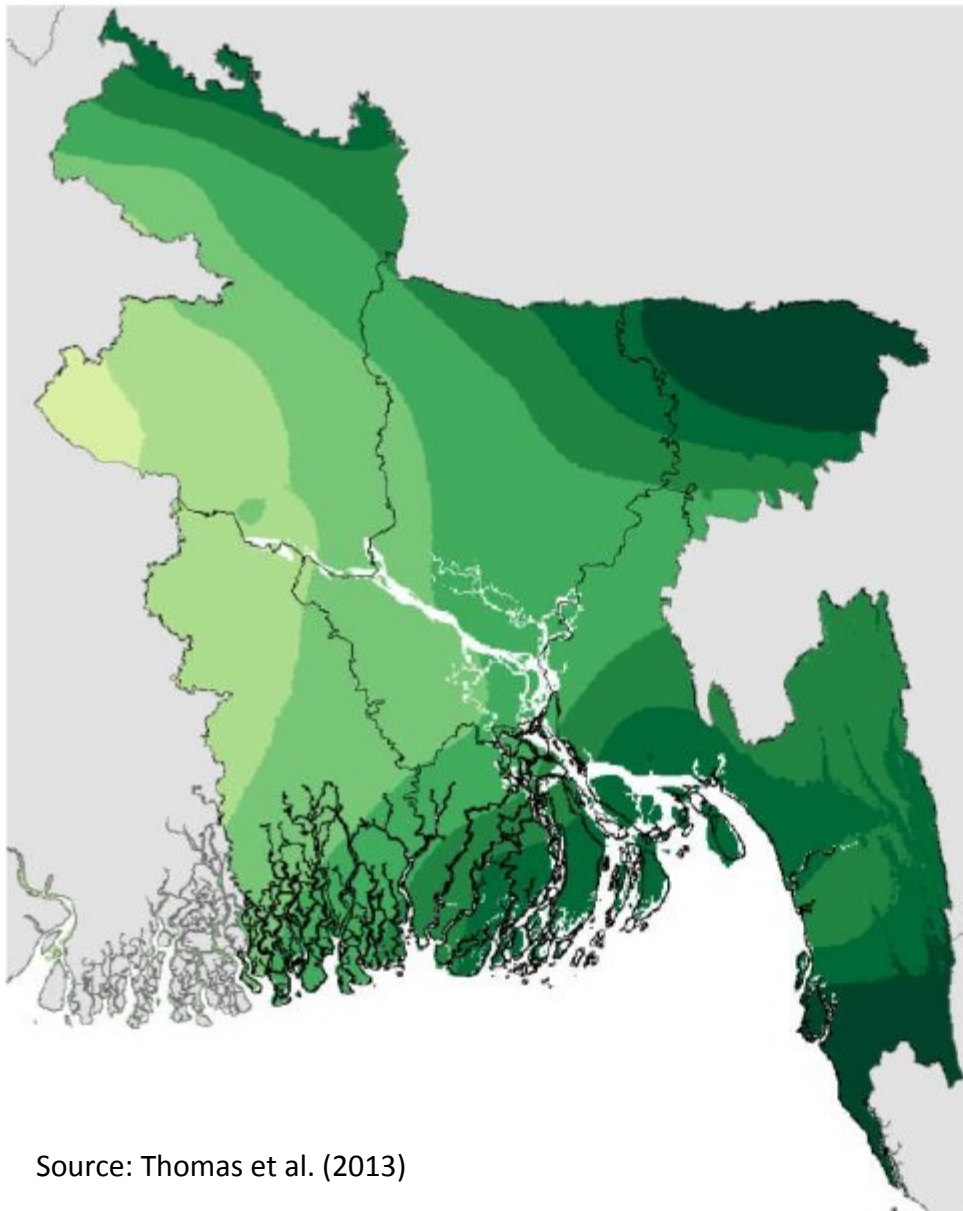


Source: Thomas et al. (2013)

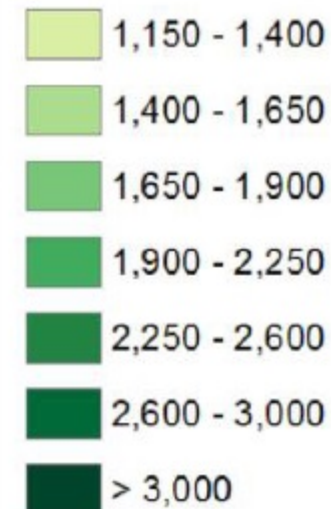


Rainfall

Rainfall in Bangladesh also differs per season and per location. The central west receives the least, less than 1,400 mm per year, while the northeast and southeast receive over 3,000 mm per year



Current average
annual rainfall
(1950-2000)

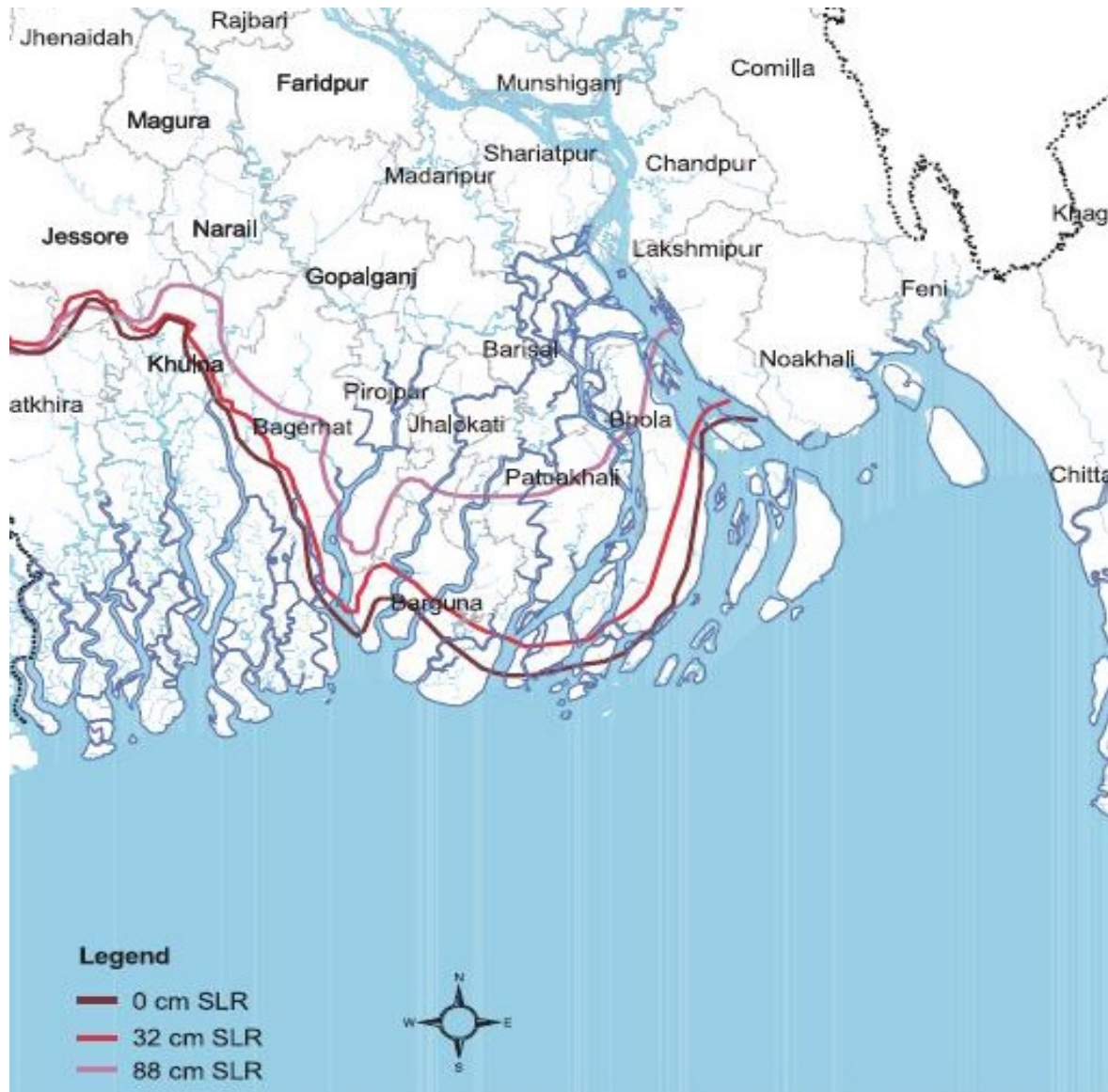


Source: Thomas et al. (2013)



Salinization

floods contribute to further salinization of coastal lands, causing not only loss of harvests but also of productive agricultural land. Out of 2.85 million hectares of coastal and offshore areas, about 1.2 million hectares of arable land are already affected by varying degrees of soil salinity



Likely salinity ingress in southern Bangladesh for different amounts of sea level rise (SLR)

Source: MoEF (2009)



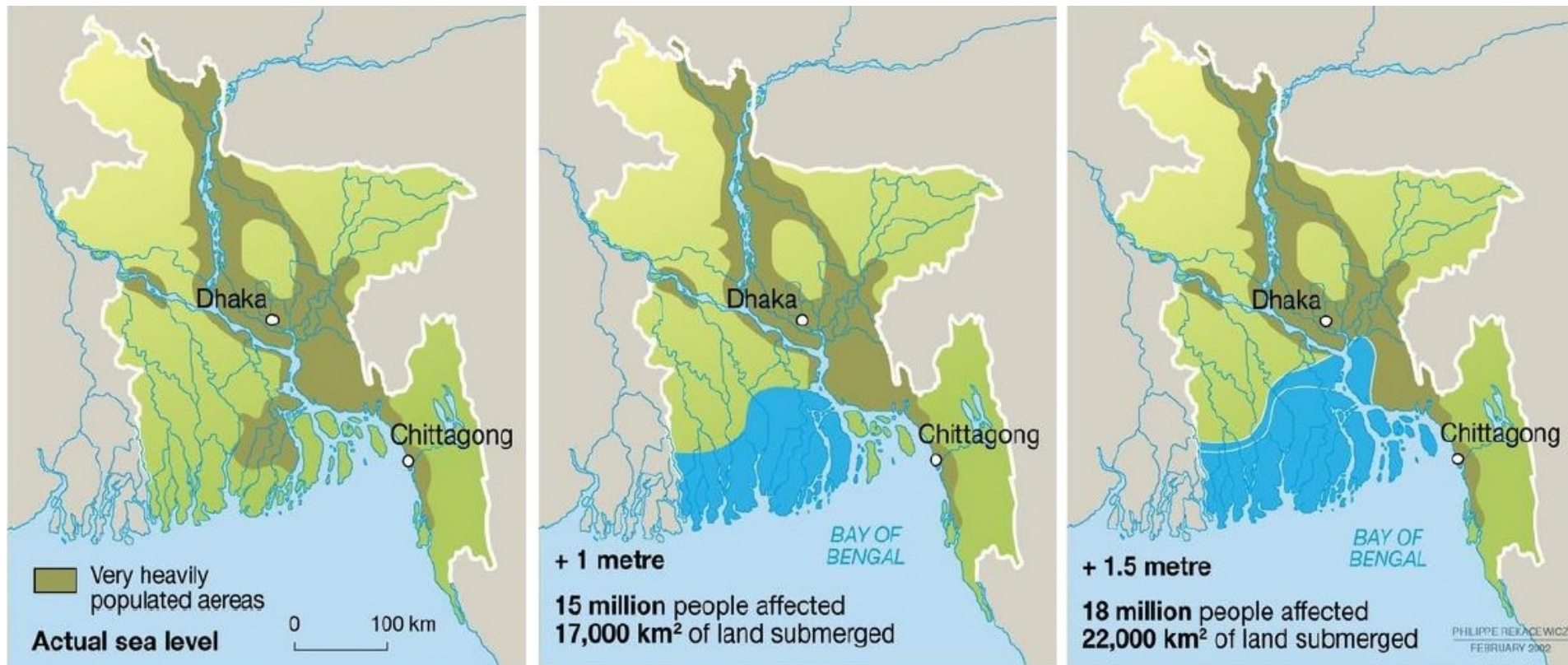
Sea level

Significant sea level rise has been measured in Bangladesh, with 4 mm per year at Hiron Point in the west, 6 mm per year at Char Changa in the centre of the country, and even 8 mm per year at Cox's Bazaar in the southeast. Sea level rise has been one of the factors that led to an increase in soil salinity in Bangladesh, from 1.5 million hectares under mild salinity in 1973 to 3 million in 2007

The IPCC projected increases of 14 cm by 2030, 32 cm by 2050, and 88 cm by 2100 (compared to 2000)



Land submerged in case of a 1.0 or 1.5 meter sea level rise





Cyclones

Due to its topography and climate, Bangladesh is subject to devastating **cyclones**, mostly in **April-May and September-November**. UNDP has ranked Bangladesh **first** of all countries in the world in terms of **vulnerability to tropical cyclones**. The country is hit by a severe cyclone on average every three years.



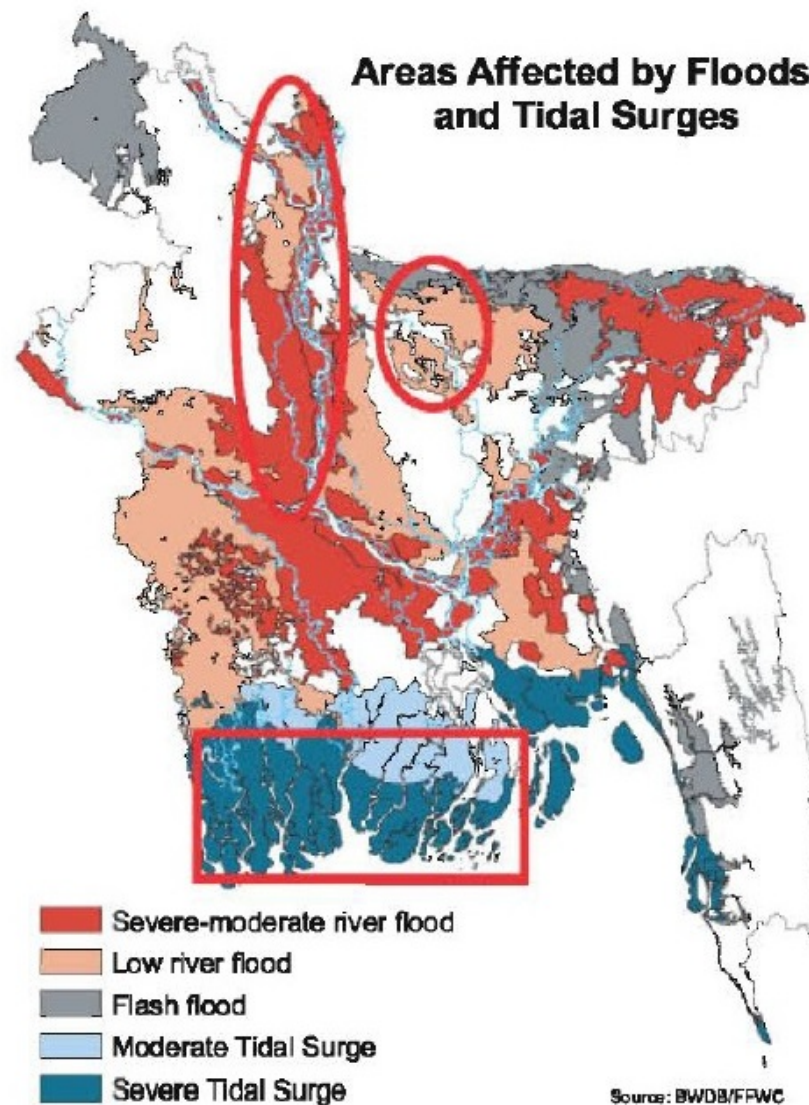
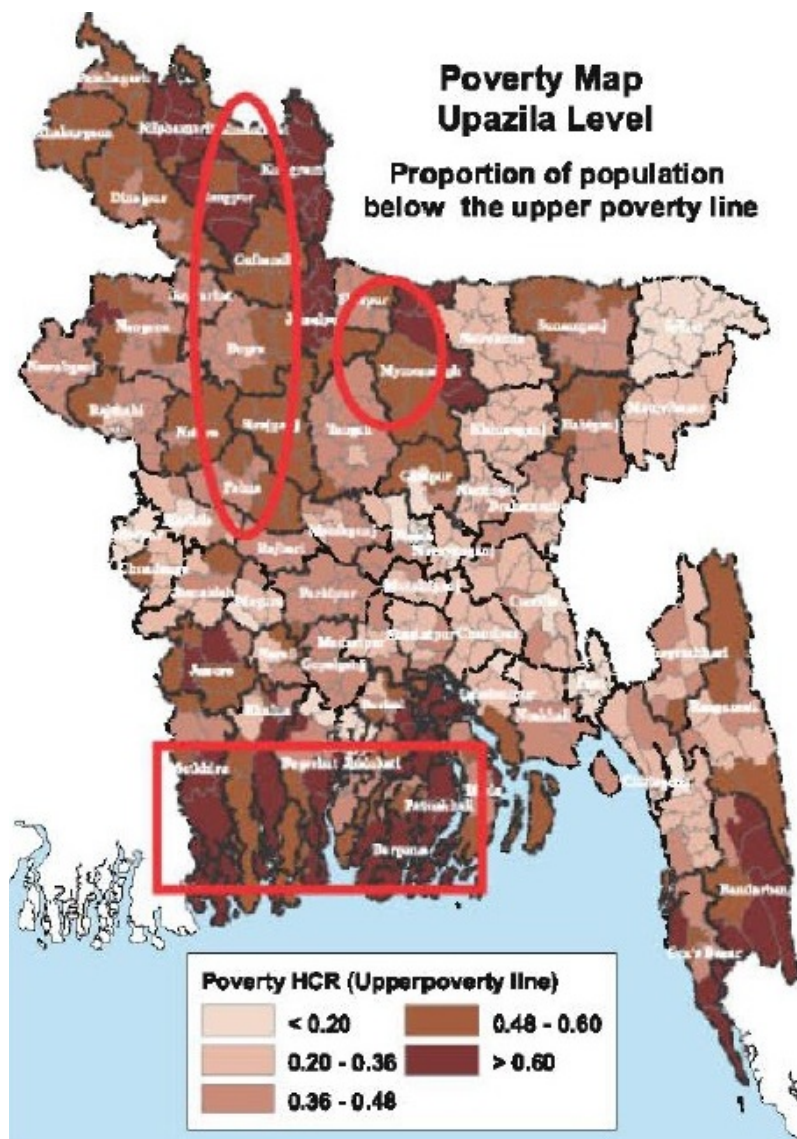
Flooding

Bangladesh is also vulnerable to flooding, with 80% of its surface forming a giant floodplain. Floods originate from precipitation in the whole of the **GBM Basin**, not just the 7% that lies within Bangladesh, and can therefore be of great magnitude. Most of the floods occur in July and August, almost every year.

These floods have devastating effects. **Riverbank erosion** results in the loss of thousands of hectares of agricultural lands and affects the population for decades.



Poverty and climate risks in the same regions



Source: BBS, World Bank, and WFP (2009):



What is important to have?

- Knowledge on environment and bio-geography
- Trans-boundary treaty
- Controlled and balanced Local Economic activity
- micro climate monitoring and remedial
- Development under National a/o Regional physical and environmental Plan
- Proper legislation and it's implementation
- Mass Awareness